

02477 Bayesian Machine Learning: Exam information (2026)

- **Formalities**

- Exam duration: 4 hours
- Date: 19/5/2025
- Aids: All aids (laptops, books, slides, code etc) allowed except AI-tools.
- ChatGPT and other AI-based tools are **not** allowed.
- There is no internet access during the exam.

- **Exam hand-in:** The hand-in for the exam is **digital**. That is, you'll need to upload a **PDF** of your final solution. However, you can also **submit an appendix on paper** if you want (e.g. if you prefer handwriting for an analytical calculation). But if you do, make sure to **reference it** from your main PDF, e.g. "Solution to Q3.2: See handwritten solution on appendix A."

- **About the digital hand-in**

- As long as your solution (text, equations etc.) is **readable, structured and easy** to follow, you can use any tool you want to produce the digital solution (e.g. Jupyter notebooks, Word, latex etc.)
- You can also copy/paste equations and figures into Word (or similar) if you prefer. Just make sure the solution as a whole is easy to read and understand.
- Make sure to test **before the exam** that you can produce a PDF without internet access.
- Jupyter notebooks can be converted to PDFs, but some students have experienced problems doing this without internet access, so make sure to test it before hand if you are planing to use this option.
- Make sure that part of the content is not "cut off" in the process of converting to PDF
- Many students prefer to convert it to HTML first and then use a browser to convert it to PDF.

- **About paper hand-in**

- Make sure to put your study number on all sheets of paper
- Write clearly

- **Type of questions:** There will be different types of questions for the exam. For example

- There will be questions on the form "Compute/determine/evaluate ... "
- There will be no "prove that"-type questions, but a few questions will require you to do **analytical calculations or reasoning**.
- There might be questions of the form: "Explain ..."

- There may be more than one way to solve some questions, e.g. by identifying and applying the correct equation or by using code from the notebooks. Both solutions are equally valid, but make sure to explain your process.
- Look at the two available exam sets for more concrete examples

- **Your solution**

- **Explain and justify your arguments** and provide intermediate results when possible
- This allows you to get full or partial credit even if you get the end result wrong (e.g. due to a typo etc)
- For each question, state the key equation(s) you are using. For questions that require you to produce a numerical answer, show code if possible or at least explain in words how you obtain the result.
- For questions requiring numeric evaluations, you are welcome to use the code from the course, but you can also use any other (offline) tools if you prefer, i.e. Python, Maple, Matlab. Just state how you arrived at your solutions.

- **Preparing for the exam**

- There is of course many ways to study for an exam, and you should of course stick to what you think is best for you. Several students have asks for suggestions for have to prepare for the exam, so here is a couple of ideas.
- If you studied and understood the material covered in the weekly exercises and the assignments, then you should be well-prepared for the exam.
- Therefore, I would suggest focusing on the exercises and the assignments, and then go to the slides or book for elaborations when needed.
- For example, for the different models we have covered, you could study
 - * what is the purpose of the model? what type of problem does it solve?
 - * how is the prior and likelihood of the model defined?
 - * how is the posterior distribution computed/approximated?
 - * how is the posterior predictive distribution computed/approximated?
 - * how do you make prediction with given the model?

- **If something is unclear**

- Let me know by email or in class, and I'll be happy to elaborate

Best of luck with the exam! :-)